

Use Case CUD Product

1. Use case code

UC234

2. Brief Description

This use case describes the way how Product Manager interacts with AIMS System to add new, view, change information or delete media products like Books, Newspapers, CDs, DVDs,.. and manually adjust inventory quantities.

3. Actors

3.1 Product Manager

4. Preconditions

- Product Manager successfully logged into the AIMS System with appropriate account provided.
- The AIMS System is currently operating normally.

5. Basic Flow of Events

1. Product Manager request access to the product management interface.
2. Software displays list of products and options for adding, editing, deleting and adjusting quantities.
3. Product Manager select Add Product
 - 3.1. Software displays general form for Product Manager
 - 3.2. Product Manager input general information and choose specific type of product.
 - 3.3. Software displays form for fill specific information depending on product type
 - 3.4. Product Manager input specific information and submit.
 - 3.5. Software checks the data and price rule.
4. Product Manager select Edit Product
 - 4.1. Product Manager select one product and choose Edit
 - 4.2. Software displays product's current information
 - 4.3. The Product Manager update the necessary information
 - 4.4. Software check constraint of data and price

5. Product Manger select Delete Product
 - 5.1. Product Manager select up to ten products to delete at a time.
 - 5.2. Product Manager confirm the delete order
 - 5.3. Software check security limit delete.
 - 5.4. Software check product quantity
 - 5.5. Software remove from sale list
6. Product Manager select adjust quantity
 - 6.1. Product Manager select product to adjust
 - 6.2. Software display box of current stock and form for adjust
 - 6.3. Product Manager input new stock and reason
7. Software display notifications and save record into history

6. Alternative flows

No	Location	Condition	Action	Resume location
1	At step 3.5 and 4.4	If the input data is not in the correct format	Software display detailed error for each data field	At step 3.4 or 4.3
2	At step 5.1	If Product Manager selects more than 10 products to delete at once.	Software displaying the warning only allows a maximum of 10 products to be selected.	At step 5.1
3	At step 5.3	If total number of products that Product Manager deleted today over 20 products.	Software notify the Product Manager of a security violation and refused to delete	At step 2
4	At step 5.4	If quantity of product removed greater than 0	Software automatic change status to deactive.	At step 7
5	At step 6.3	If Product Manager do not input Reason field	Software require to enter reason for adjust product quantity	At step 6.3
6	Any step	If Product Manager cancel process	Software do not save any changes and return product list interface	At step 2

7. Input data

No	Data fields	Description	Mandatory	Valid condition	Example
1	Barcode	Unique identifier for the product	Yes	Unique in the system; numeric only	893123456789
2	Title	The name of the media product	Yes	Cannot be blank	DVD Hello World
3	Category	Product classification	Yes	One of: Books, Newspapers, CDs, DVDs	Books
4	General Description	Condition, primary color, or return condition	Yes	Text description	New condition, blue cover
5	Original Value	Base price excluding 10% VAT	Yes	Positive integer	100
6	Current Price	Selling price excluding 10% VAT	Yes	Between 30% and 150% of Original Value	120
7	Dimensions	Physical size (Height x Width x Length)	Yes	Positive values only	20 x 15 x 3 (cm)
8	Weight	Physical weight of the product	Yes	Positive value (kg)	0.5
9	Author	Creator(s) of the book	Yes (Book)	Required if Category is "Books"	Ngo Tat To
10	Cover Type	Format of the book cover	Yes (Book)	"Paperback" or "Hardcover"	Paperback
11	Editor-in-chief	Head of the newspaper	Yes (News)	Required if Category is "Newspapers"	Nguyen Van A
12	Artists	Performers or singers	Yes (CD)	Required if Category is "CDs"	Son Tung M-TP
13	Track List	List of songs with titles and lengths	Yes (CD)	Must include title and duration per track	Track 1 - 10:30
14	Disc Type	Physical format of the disc	Yes (DVD)	"Blu-ray" or "HD-DVD"	Blu-ray
15	Director	Director of the media	Yes (DVD)	Required if Category is "DVDs"	Christopher Nolan
16	New Stock	Updated manual inventory count	Yes (Adjust)	Non-negative integer	50
17	Reason	Explicit record for manual stock change	Yes (Adjust)	Required for all manual adjustments	Damaged during transit

8. Output data

No	Data fields	Description	Display format	Example
1	Operation Status	Notification of the result for the performed action (add, edit, delete, or adjust).	Text notification	Product added successfully
2	Product List	A list of all media products currently available in the store's inventory.	Data table	(List of Books, CDs, ...)
3	Product Details	Full detailed information of a specific product when viewed or being edited.	Form view	Title: Prison Break, Price: 120,000,...
4	Product Status	The commercial availability of the product (available or deactivated).	Label/Tag	Deactivated
5	History Log	A record list of all product additions, edits, and deletions.	Timeline/Log entry	16/03/2026: Manager A updated stock of Barcode X
6	Error Message	Detailed notification when input data is invalid or violates system rules.	Text (Red)	Price must be between 30% and 150% of Original Value
7	Stock Quantity	The actual number of items in stock after a manual update or a sale.	Positive integer	45
8	Adjustment Reason	An explicit record explaining why a manual stock change was performed.	Text description	Damaged during transit

9. Postconditions

- Stored data: All products successfully saved into database or new information be updated with newest version.
- Inventory Accuracy: The stock quantity of product is accurately updated in system according to manual adjustment
- Product status: If product with zero stock was deleted, removed it from sales list, if its stock greater than zero, its status changed to Deactivated.
- Audit trail: A new record created in history, log the detail information of the operation
- Stock reason logging: Explicit reason is saved and linked to stock change record
- User notification: Product Manager receive notification from software, product list interface must refreshed to display updated list